

Question	Answer	Marks
6(a)	(motion in which) acceleration is (directly) proportional to displacement	B1
	(motion in which): <i>either:</i> acceleration is (always) in the opposite <u>direction</u> to displacement <i>or:</i> acceleration is (always) <u>directed</u> towards a fixed point	B1
6(b)(i)	$\omega = 2\pi / T$ <u>and</u> ω identified as 6.9	C1
	$T = 2\pi / 6.9$ $= \mathbf{0.91\ s}$	A1
6(b)(ii)	$v_0 = \omega x_0$	C1
	$= 6.9 \times 3.2 = 22\ \text{mm s}^{-1}$	A1
6(c)(i)	$E = BA / t$	C1
	$E_0 = 0.047 \times 2 \times 0.060 \times (22 \times 10^{-3} \Delta t) / \Delta t$	C1
	$= \mathbf{1.2 \times 10^{-4}\ V}$	A1
6(c)(ii)	<u>sketch:</u> two cycles of a sinusoidal curve of period T and amplitude E_0	B1
	sinusoidal curve of correct phase, with $E = 0$ at (and only at) $t = 0.25T$, $t = 0.75T$, $t = 1.25T$ and $t = 1.75T$	B1